## V. HOW PLATFORM ECONOMICS AND CAPITAL MARKETS MAY FACILITATE ANTICOMPETITIVE CONDUCT AND STRUCTURES

As Part IV mapped out, aspects of Amazon's conduct and structure may threaten competition yet fail to trigger scrutiny under the analytical framework presently used in antitrust. In part this reflects the "consumer welfare" orientation of current antitrust laws, as critiqued in Part II. But it also reflects a failure to update antitrust for the internet age. This Part examines how online platforms defy and complicate assumptions embedded in current doctrine. Specifically, it considers how the economics and business dynamics of online platforms create incentives for companies to pursue growth at the expense of profits, and how online markets and control over data may enable new forms of anticompetitive activity.

Economists have analyzed extensively how platform markets may pose unique challenges for antitrust analysis.[377] Specifically, they stress that analysis applicable to firms in single-sided markets may break down when applied to two-sided markets, given the distinct pricing structures and network externalities.[378] These studies often focus on the challenge that two-sided platforms face in attracting both sides—the classic coordination problem of having to attract buyers without an established line of sellers, and vice versa.[379] Economists tend to conclude that—given the particular challenges of two-sided markets[380]—antitrust should be forgiving of conduct that might otherwise be characterized as anticompetitive.[381]

Legal analysis of online platforms is comparatively undertheorized. The Justice Department's case against Microsoft under Section 2 of the Sherman Act, initiated in the 1990s, remains the government's most significant case involving two-sided markets—even as platforms have emerged as central arteries

377. *See* David S. Evans, *The Antitrust Economics of Multi-Sided Platform Markets*, 20 YALE J. ON REG. 325 (2003); King, *supra* note 376; David S. Evans & Richard Schmalensee, *The Antitrust Analysis of Multi-Sided Platform Businesses* (Coase-Sandor Inst. for Law & Econ., Working Paper No. 623, 2012).

378. *See* Julian Wright, *One-Sided Logic in Two-Sided Markets*, 3 REV. NETWORK ECON. 44 (2004).

379. *See* DAVID S. EVANS, PLATFORM ECONOMICS: ESSAYS ON MULTI-SIDED BUSINESS 112 (2011).

380. Two-sided markets are platforms that have two distinct user groups that offer each other network benefits.

381. *See* EVANS, *supra* note 379, at 112 ("The pricing and investment strategies that firms in two-sided markets use to 'get both sides on board' and 'balance the interests of both sides' raise novel ones. These pricing and other business strategies are needed to solve a fundamental economic problem arising from the interdependency of demand on both sides of the market. In some cases, the product could not even exist without efforts to subsidize one side of the market or the other.").

in our modern economy. Starting in 2011, the FTC pursued an investigation into Google, partly in response to allegations that the company uses its dominance as a search engine to cement its advantage and exclude rivals in other lines of business. While the FTC closed the investigation without bringing any charges, leaks later revealed that FTC staff had concluded that Google abused its power on three separate counts.[382] The European Union has brought charges against Google for violating antitrust laws.[383]

For the purpose of competition policy, one of the most relevant factors of online platform markets is that they are winner-take-all. This is due largely to network effects and control over data, both of which mean that early advantages become self-reinforcing. The result is that technology platform markets will yield to dominance by a small number of firms. Walmart's recent purchase of the one start-up that had sought to challenge Amazon in online retail—Jet.com—illustrates this reality.[384]

Network effects arise when a user's utility from a product increases as others use the product. Since popularity compounds and is reinforcing, markets with network effects often tip towards oligopoly or monopoly.[385] Amazon's user reviews, for example, serve as a form of network effect: the more users that have purchased and reviewed items on the platform, the more useful information other users can glean from the site.[386] As the Fourth Circuit has noted, "[O]nce dominance is achieved, threats come largely from outside the dominated market, because the degree of dominance of such a market tends to become so extreme."[387] In this way, network effects act as a form of entry barrier. 시장 진입 장벽

A platform's control over data, meanwhile, can also entrench its position.[388] Access to consumer data enables platforms to better tailor services and gauge demand. Involvement *across* markets, meanwhile, may permit a company to use

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**382.** Brody Mullins et al., *Inside the U.S. Antitrust Probe of Google*, WALL ST. J. (Mar. 19, 2015, 7:38 PM), http://www.wsj.com/articles/inside-the-u-s-antitrust-probe-of-google-1426793274 [http://perma.cc/H4PZ-JZ9K].

**383.** Mark Scott & James Kanter, *Google Faces New Round of Antitrust Charges in Europe*, N.Y. TIMES (July 14, 2016), http://www.nytimes.com/2016/07/15/technology/google-european-union-antitrust-charges.html [http://perma.cc/2SYP-5Z4B].

**384.** *See supra* note 319.

**385.** STUCKE & GRUNES, *supra* note 47, at 163.

**386.** This is a form of "scale of data" network effect rather than a "traditional network effect." *Id.* at 170.

**387.** Novell Inc. v. Microsoft Corp., 505 F.3d 302, 308 (4th Cir. 2007).

**388.** *See* Guy Rolnik & Asher Schechter, *Is the Digital Economy Much Less Competitive than We Think It Is?*, PROMARKET (Sept. 23, 2016), http://promarket.org/digital-economy-much-less-competitive-think [http://perma.cc/K2R6-TB7Q].

data gleaned from one market to benefit another business line.[389] Amazon's use of Marketplace data to advantage its retail sales, as described in Section IV.D, is an example of this dynamic. Control over data may also make it easier for dominant platforms to enter new markets with greater ease. For example, reports now suggest that Amazon may dramatically expand its footprint in the ad business, "leveraging its rich supply of shopping data culled from years of operating a massive e-commerce business."[390] In other words, control over data, too, acts as an entry barrier.

이미 우위를 점한 기업은
쉽게 새로운 확장 가능 (데이터 통제 여력)

Given that online platforms operate in markets where network effects and control over data solidify early dominance, a company looking to compete in these markets must seek to capture them. The most effective way is to chase market share and drive out one's rivals—even if doing so comes at the expense of short-term profits, since the best guarantee of long-term profits is immediate growth. Due to this dynamic, striving to maximize market share at the expense of one's rivals makes predation highly rational; indeed, it would be irrational for a business *not* to frontload losses in order to capture the market. Rec-Recognizing that enduring early losses while aggressively expanding can lock up a monopoly, investors seem willing to back this strategy.

단기 이익이 줄어들더라도
장기를 위해 시장점유율을
빠르게 확보하는 것이 중요
↓
초기 손실 감수 + 공격적 확장이
장기적 독점 확보로 이어질 수 있기 때문

As the Introduction and Part III describe, Amazon has charted immense growth while investing aggressively—both by expanding provision of physical and online infrastructure and by pricing goods below cost. Amazon's stock price has soared despite a history of razor-thin—or even negative—margins. In essence, investors have given Amazon a free pass to grow without any pressure to show profits. The firm has used this edge to expand wildly and dominate online commerce.

<아마존 사례>
상품을 원가 이하로 판매 → 공격적 확장
(이후 독점 달성 시 손실을 회수할 수
있을 것이라는 믿음 O)

The idea that investors are willing to fund predatory growth in winner-take-all markets also holds in the case of Uber. Although the dynamics of the online retail market are distinct from those of ride-sharing, Uber's growth trajectory is worth analyzing for general insight into how investors enable plat-

389. Interestingly, agencies have required vertically merging parties to erect firewalls to prevent anticompetitive use of data. *See, e.g.*, *In re* Coca-Cola Co., 150 F.T.C. 520, 2010 WL 9549986 (2010) (ordering Coca-Cola to set up a firewall to ensure that its merger with a bottling subsidiary does not give it access to information from its competitor, Dr. Pepper Snapple Group); Press Release, FTC, FTC Puts Conditions on Coca-Cola's $12.3 Billion Acquisition of its Largest North American Bottler (Sept. 27, 2010), http://www.ftc.gov/news-events/press-releases/2010/09/ftc-puts-conditions-coca-colas-123-billion-acquisition-its [http://perma.cc/BP7U-EY33] (discussing the Coca-Cola settlement and a similar PepsiCo settlement).

390. Mike Shields, *Amazon Looms Quietly in Digital Ad Landscape*, WALL ST. J. (Oct. 6, 2016, 3:28 PM), http://www.wsj.com/articles/amazon-looms-quietly-in-digital-ad-landscape-1475782113 [http://perma.cc/5ACL-MJ7D].

form dominance. In 2015, news reports revealed that Uber had an operating loss of $470 million on $415 million in revenue, confirming suspicions that the company has been bleeding money for the sake of achieving steep growth and acquiring market share.[391] In China, the company has lost more than $1 billion a year.[392] The strategy of aggressive price competition and brazen leadership coupled with soaring growth prompted immediate comparisons to Amazon.[393] Like Amazon, Uber has drawn immense interest from investors. As of July 2015, its valuation hit nearly $51 billion, equaling the record set by Facebook in 2012.[394] It recently secured an additional $3.5 billion in investment, bringing its total funds to $13.5 billion—a figure "far greater than most companies raise even during an initial public offering," which Uber has avoided.[395]

One might dismiss this phenomenon as irrational investor exuberance. But another way to read it is at face value: the reason investors value Amazon and Uber so highly is because they believe these platforms will, eventually, generate huge returns. As one venture capitalist recently remarked, if he had to "put his entire capital in a single company and hold it for the next 10 years," he would choose Amazon. "I don't see any cleaner monopoly available to buy in the public markets right now."[396] In other words, that these platform companies are

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**391.** *See* Eric Newcomer, *Uber Draws Fresh Amazon Comparisons as Growth Trumps Profit*, BLOOMBERG (July 1, 2015, 12:30 AM), http://www.bloomberg.com/news/articles/2015-07-01/uber-draws-fresh-comparison-with-amazon-as-growth-trumps-profit [http://perma.cc/AYF9-9RJ7]. Uber does not just lose money in the aggregate by reinvesting more than it generates, but also by pricing rides below what it pays drivers. In other words, it is pricing below its variable costs—which enforcers traditionally read as a sign of predatory pricing. "As anyone who has taken an Uber and talked to the driver knows, sometimes the fare collected from the rider is less than what Uber pays the driver." *Id.*

**392.** Charles Clover & Leslie Hook, *Uber Losing More than $1bn a Year in China*, FIN. TIMES (Feb. 18, 2016, 7:25 PM), http://www.ft.com/content/f889f812-d664-11e5-829b-8564e7528e54 [http://perma.cc/6U6P-JQ7Q].

**393.** "'They're wise to expand as fast as they can,' said Lou Shipley, a lecturer at the MIT Sloan School of Management. 'I would liken it to what Amazon did with books.'" Newcomer, *supra* note 391.

**394.** Douglas MacMillan & Telis Demos, *Uber Valued at More than $50 Billion*, WALL ST. J. (July 31, 2015, 8:50 PM), http://www.wsj.com/articles/uber-valued-at-more-than-50-billion-1438367457 [http://perma.cc/T6GW-SY2J].

**395.** Leslie Hook, *Uber Cranks Up Ride-Hailing Battle with $3.5 bn Saudi Investment*, FIN. TIMES (June 2, 2016), http://www.ft.com/content/3ac7c982-2879-11e6-8b18-91555f2f4fde [http://perma.cc/RXV8-TPBP].

**396.** Eugene Kim, *Billionaire VC Says that Most Companies Will Eventually Pay an Amazon 'Tax,'* BUS. INSIDER (Jan. 21, 2016, 4:21 AM), http://www.businessinsider.in/Billionaire-VC-says-that-most-companies-will-eventually-pay-an-Amazon-tax/articleshow/50662558.cms [http://perma.cc/4ZGS-VSL7].

undertaking consistent, steep losses and still generating strong investor backing suggests that the markets expect Amazon and Uber to recoup these losses.

While investors have unambiguously endorsed and funded online platforms' quest to bleed money in their race to draw users, antitrust doctrine fails to acknowledge this strategy. In the past, the Supreme Court's analysis has embraced the Efficient Market Hypothesis (EMH), the idea that market prices reflect all available information.[397] The Justice Department also acknowledges that market information—for example, the financial terms of an acquisition—may "be informative regarding competitive effects."[398] Applying EMH in this instance overwhelmingly suggests that these platforms are positioned to recoup their losses. Yet bringing a predatory pricing suit against an online platform would be almost impossible to win in light of the recoupment requirement. Strikingly, the market is reflecting a reality that our current laws are unable to detect.[399] 시장과 법의 괴리

In addition to overlooking why online platform dynamics make predation especially rational, current doctrine also fails to appreciate how a platform might recoup losses. For one, investor support allows Amazon to strategize and operate on a time horizon far longer than what the *Brooke Group* or *Matsushita* Courts confronted. Raising prices in a third year after enduring losses for two is different from engaging in a decade-long quest to become the dominant online retailer and provider of internet infrastructure. That longer timeline, meanwhile, makes available more recoupment mechanisms. Not only has Amazon inaugurated an entire generation into online shopping through its platform, but it has expanded into a suite of additional businesses and amassed significant troves of data on users. This data enables it both to extend its tug over customers through highly tailored personal shopping experiences, and, potentially, to institute forms of price discrimination, as described in Section IV.A. Both the latitude granted by investors and control over data equip an in-

397. The Supreme Court has affirmed the validity of EMH. *See* Halliburton Co. v. Erica P. John Fund, 134 S. Ct. 2398, 2409-11, 2417 (2014).

398. *Horizontal Merger Guidelines*, *supra* note 44, at 4 ("For example, a purchase price in excess of the acquired firm's stand-alone market value may indicate that the acquiring firm is paying a premium because it expects to be able to reduce competition or to achieve efficiencies.").

399. Ironically, the logic that is motivating investors—the idea that it is worth encouraging platforms to bleed money to establish a dominant position and capture the market, at which point these firms will be able to recoup those losses—maps on to the logic underpinning current predatory pricing doctrine. The main issue is how narrowly the law currently conceives of recoupment, which does not account for how Amazon can leverage its multiple lines of business.

cumbent platform to recoup losses in ways less obviously connected to the initial form of below-cost pricing.

These recoupment mechanisms may also be more sophisticated than what a judge or even rivals would be able to spot. This last point becomes even more apparent in the context of Uber, whose dynamic pricing has conditioned users not to expect a stable or regular price. While Uber claims that its algorithms set prices to reflect real-time supply and demand, initial research has found that the company manipulates the availability of both.[400] Moreover, it routinely gives away discount coupons to select users, effectively charging users different prices, even for the same service at the same time.[401]

Although platforms form the backbone of the internet economy, the way that platform economics implicates existing laws is relatively undertheorized.[402] Amazon's conduct suggests that predatory pricing and integration across related business lines are emerging as key paths to establishing dominance—aided by the control over data that dominant platforms enjoy. But because current predatory pricing doctrine defines recoupment in overly narrow terms, competitors generally have not been able to make an effective legal case. Similarly, because current doctrine largely discounts entry barriers, the anticompetitive effects of vertical integration are difficult to cognize under the existing framework. Roadblocks to these claims persist even as Amazon's valuation and share price point to a strong market expectation of recoupment and profits.

법이 '손실 회수'를 좁게 정의
→ 경쟁자들이 소송 제기하기 어렵다
ex) 이미 시장에서는 우버의 초저 손실에도 불구하고 장기적 수익 창출의 기대가 형성됨
→ 그러나, 법은 이를 포착하지 못한다

There are signs that enforcers are becoming more attuned to the special factors that may render current antitrust analysis inadequate to promote competition in internet platform markets. For example, in 2014 the United States successfully challenged a merger between two leading providers of online ratings and reviews platforms. In its complaint, DOJ acknowledged that data-driven industries can be characterized by network effects, which increase switching costs and entry barriers.[403] Recent comments by FTC Commissioner Terrell

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400. *See* Tim Hwang & Madeleine Clare Elish, *The Mirage of the Marketplace: The Disingenuous Ways Uber Hides Behind Its Algorithm*, SLATE (July 27, 2015, 6:00 AM), http://www.slate.com/articles/technology/future_tense/2015/07/uber_s_algorithm_and_the_mirage_of_the_marketplace.html [http://perma.cc/B5UR-P9PN].

401. *See* Felix Salmon, *Why the Internet Is Perfect for Price Discrimination*, REUTERS (Sept. 3, 2013), http://blogs.reuters.com/felix-salmon/2013/09/03/why-the-internet-is-perfect-for-price-discrimination [http://perma.cc/NZ4E-SVJJ].

402. *See* David Singh Grewal, *Before Peer Production: Infrastructure Gaps and the Architecture of Openness in Synthetic Biology*, 20 STAN. TECH. L. REV. (forthcoming 2017).

403. The Justice Department wrote, "[A]s more retailers purchase Bazaarvoice's PRR platform, the Bazaarvoice network becomes more valuable for manufacturers because it will allow[ ] them to syndicate content to a greater number of retail outlets. The feedback between the manufacturers and retailers creates a network effect that is a significant and durable compet-

McSweeny—noting that data can act as a barrier to entry and that "competition enforcers can and should assess the competitive implications of data"—also suggest that top officials are assessing how to revise their tools and framework for gauging competition in platform markets.[404]

While this burgeoning recognition is heartening, the unique features of platform markets require a more thorough evaluation of how antitrust is applied. Because scale is both vital to platforms' business model and helps entrench their dominant position, antitrust should reckon with the fact that pursuing growth at the expense of returns is—contra to current doctrine—highly rational. An approach more attuned to the realities of online platform markets would also recognize the variety of mechanisms that businesses may use to recoup losses, the longer time horizon on which recoupment might occur, and the ways that vertical integration and concentrated control over data may enable new forms of anticompetitive conduct. Revising antitrust to reflect the dynamics of online platforms is vital, especially as these companies come to mediate a growing share of communications and commerce.

반독점법 개정 필요
: 현행은 가격 인상 여부+단기적 기준을
적용하여 장기적으로 이루어진 플랫폼 독점을
제대로 규제하지 못한다

## VI. TWO MODELS FOR ADDRESSING PLATFORM POWER

If it is true that the economics of platform markets may encourage anticompetitive market structures, there are at least two approaches we can take. Key is deciding whether we want to govern online platform markets through competition, or want to accept that they are inherently monopolistic or oligopolistic and regulate them instead. If we take the former approach, we should reform antitrust law to prevent this dominance from emerging or to limit its scope. If we take the latter approach, we should adopt regulations to take advantage of these economies of scale while neutering the firm's ability to exploit its dominance.

### *A. Governing Online Platform Markets Through Competition*

Reforming antitrust to address the anticompetitive nature of platform markets could involve making the law against predatory pricing more robust and strictly policing forms of vertical integration that firms can use for anticompeti-

itive advantage for Bazaarvoice." Complaint at 18, United States v. Bazaarvoice, Inc., No. 13-0133 2014 (N.D. Cal. Jan. 10, 2013), 2014 WL 203966.

404. Terrell McSweeny, Comm'r, FTC, Remarks to the U.S. Chamber of Commerce at TecNation 2016 (Sept. 20, 2016), http://www.ftc.gov/system/files/documents/public_statements/985773/mcsweeny_-_tecnation_2016_9-20-16.pdf [http://perma.cc/N7GA-YN5P].